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Young and old users prefer immersive virtual reality over a social robot for short-term cognitive training

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ABSTRACT

Socially Assistive Robots (SARs) and immersive Virtual Reality (iVR) are interactive platforms that promote user engagement, which can motivate users to adhere to therapeutic frameworks. SARs use social presence to create affective relationships with users, leveraging the human tendency to be driven by social interactions. iVR uses spatial presence to provide an intense multisensory experience that submerges users in a virtual world. We adapted two such platforms – a SAR and an iVR – to deliver cognitive training (CT), by integrating established cognitive tasks in gamified environments that convey a strong sense of presence. Sixty-four participants underwent CT with both platforms. We tested: (1) their perception of both platforms; (2) whether they preferred one over the other in the short term; (3) their projected preferences for long-term training; and (4) whether their preferences correlated with personal characteristics. They preferred the virtual experience in the short term across age and gender. For long-term CT, there was equal projected preference for both platforms. It may be that a combination of social and spatial presence might yield engagement in long-term training.

1. Introduction

Overview

User interfaces aimed at therapeutic purposes often include an element of either social or spatial presence to increase user engagement (Abdi et al., 2018; Darekar et al., 2015; Dias et al., 2019; Feil-Seifer and Mataric, 2005; Ghai and Ghai, 2019; Kandalaft et al., 2013; Maples-Keller et al., 2017; Pennisi et al., 2016; Porras et al., 2018; Tapus et al., 2009b; Van der Loos et al., 2016; Weiss et al., 2014). As novel interactions with technology are developed for long-term engagement in a therapeutic context, it is important to understand the differences in users' perceptions of these two types of presence, which may change depending on the users' profile and on the specific context. In this work, we examine these differences in the context of cognitive training: we use two different technological platforms to invoke a sense of presence. We therefore start with a brief overview of what is presence, and then introduce the topic of cognitive training, the associated difficulty in keeping people engaged in training over time, and solutions that have been previously attempted; we then describe the two types of platforms that we used in this work – a socially assistive robot and immersive virtual reality, which we used to elicit social presence and spatial presence, respectively; we conclude with the specific aims of this work.

> "Designing a user-centric interface means using social and spatial cues for computer users so they can interact with user *interfaces in the same ways they interact with the world in everyday life […] Two forms of presence are key to this effort, social presence and spatial presence*" (Horvath and Lombard, 2010).

Presence

The concept of *social presence* refers to experiencing an artificially simulated social actor as an actual social entity (Lee, 2004), while the concept of *spatial presence* refers to the perceived experience of being physically present in a non-physical world (Lee, 2004). It can be simply described in everyday language as "the sense of being there," where one is not watching the environment, but rather participating in it (e.g., Biocca and Delaney, 1995; Jung and Lee, 2004; Lombard and Ditton, 1997; Slater, 2009; Wirth et al., 2007); for a review of this topic, see

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Cummings and Bailenson)2016).

Cognitive training

Cognitive training is a field of research and clinical interventions in which engagement and motivation are essential for effective outcomes (Coyle et al., 2015; Kueider et al., 2012; Lumsden et al., 2016). As the aging population grows rapidly, changes in cognitive function of older people become a major concern. A large body of research associates age with deficits in executive functions, which were found to lead to difficulties in performing daily activities and in social functioning (Cohen--Zimerman and Hassin, 2018; Glisky, 2007; Salthouse, 1996; Wilson et al., 2013). Various methods for maintaining cognitive function, such as cognitive training (Anguera et al., 2013; Ball et al., 2007; Kueider et al., 2012), physical activity (Angevaren et al., 2008), and social activities (James et al., 2011), indicate that the decline can be slowed down (Shah et al., 2017).

Despite its demonstrated effectiveness, cognitive training is considered effortful and frustrating to perform (Lumsden et al., 2016), thereby reducing its effectiveness and increasing drop-out rates during training (Lumsden et al., 2016). Since cognitive training requires many repetitions and adherence (Lumsden et al., 2016; Park and Bischof, 2013; Shah et al., 2017), it is necessary to find engaging tools that would keep users interested and motivated (Park and Bischof, 2013). This is particularly important for individuals for whom cognitive training in a community setting is not available, such as is the case in the midst of the COVID-19 pandemic for many older adults, who are in a high-risk group, and therefore adhere to social distancing rules and avoid public gatherings. To increase user interest and motivation, previous works have focused on *gamification* of the training, by inserting elements such as a score, leaderboards, and a narrative (for reviews, see Kueider et al., 2012; Lumsden et al., 2016). Here, we focus on two technologies that can be harnessed to help gamify cognitive training: socially assistive robots, and immersive virtual reality; we briefly review these below.

Socially assistive robots

Socially assistive robots (SARs) are a combination of two types of robots: *assistive robots*, which assist people with physical disabilities, and *socially interactive robots*, which interact and communicate with humans. The goal of SARs is to create close and affective interactions with human users for the purpose of giving assistance and achieving progress in physical and mental functioning (Feil-Seifer and Mataric, 2005). SARs use the impact of social interaction as a tool for assistance in a variety of contexts, such as cognitive training (Kim et al., 2015; Law et al., 2019a; Manca et al., 2020; Tapus et al., 2009b), exercise (Eizicovits et al., 2018; Fitter et al., 2020; Kashi and Levy-Tzedek, 2018), senior care (Abdi et al., 2018), dementia care (Broadbent et al., 2018; Darragh et al., 2017;

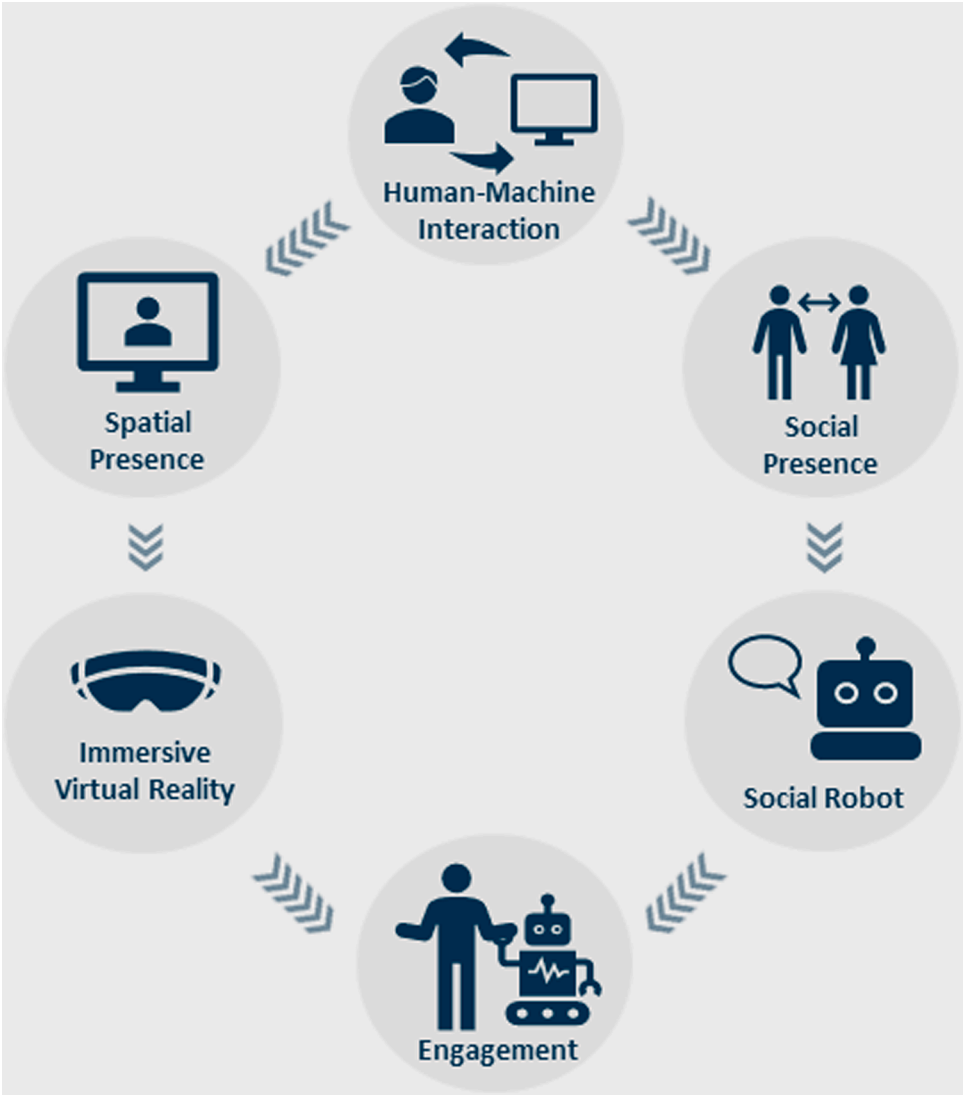


Fig. 1. Conceptual map of the strengths of each of the two platforms, leading to user engagement. The diagram illustrates two different ways in which user engagement is achieved in human-machine interaction: either through social presence – which we implemented by using a social robot – or through spatial presence – which we implemented by using immersive virtual reality. Both types of presence leverage users' motivational mechanisms, which, in turn, yield engagement and can increase adherence in therapeutic frameworks.

Law et al., 2019b), rehabilitation (Feingold Polak and Tzedek, 2020; Feingold-Polak et al., 2021; Van der Loos et al., 2016), pain management (Geva et al., 2020) and social-behavior tutoring for children with autism (Pennisi et al., 2016).

A core element that differentiates SARs from other automated assistive technologies is their *social presence* (see Fig. 1). Social robots exhibit social cues and human characteristics in order to communicate and elicit social responses, in turn leading to greater enjoyment and social influence (Oh et al., 2018; Riek et al., 2009). The anthropomorphic shape of a robot – its resemblance to a human form – as well as its human-like behavior, improves human-robot communication (Fink, 2012), corresponding to the assumption that people prefer to communicate with automated entities in a way the resembles communication with other humans (Fong et al., 2003). Robots that possess human-like qualities exploit the natural tendency of humans to attribute human characteristics and emotional states to artificial entities in order to rationalize and understand their behavior in social environments (Duffy, 2003). Additional social characteristics of the robot, such as empathy (Leite et al., 2013), sense of humor (Tay et al., 2016), and personality traits (Tapus and Matarić, 2008), have been found to enhance the robot's *social presence*. Considering the subjective experience that social communication can elicit, users can perceive interaction with a social robot in different ways according to interpersonal differences among users, such as age (De Graaf and Allouch, 2013), gender (Kuo et al., 2009), and personality traits (Oh et al., 2018). The users' perceptions of the robot vary depending on whether they trust the robot to fulfill the role with which it was charged in a particular context (Kellmeyer et al., 2018; Langer et al., 2019), and their interaction with it may affect their interactions with people around them (Langer and Levy-Tzedek, 2021).

Virtual reality

Virtual Reality (VR) generally refers to a computer-mediated simulation, imitating a physical environment (Slater and Sanchez-Vives, 2016). The user can navigate through an imaginary setting and interact with items or other virtual entities, using either body movements or a remote controller, artificially creating a sensory experience of being inside a virtual world. Over the years, VR has been implemented in many forms of hardware, with a head-mounted display (HMD) being a recent implementation of VR, widely used today.

In recent years, VR technology has been widely used in clinical research. Clinical fields utilize VR as a motivational tool to get patients engaged in performing their required training (e.g. Dias et al., 2019; Porras et al., 2018). The option to create customized reality enables the development of therapeutic tools for both psychological and physical impairments such as post-stroke rehabilitation (Darekar et al., 2015), post-traumatic stress disorder and phobias (Maples-Keller et al., 2017), cerebral palsy (Ghai and Ghai, 2019), autism (Kandalaft et al., 2013), and motor deficits (Weiss et al., 2014). When engaged in a VR activity, the experience can be immersive or not.

Immersion

The concept of *immersion* is often confused with the concept of *presence*; however, the first refers to the quality of the VR system, and the second to the subjective experience of the user. Biocca & Delaney described *immersion* as "the degree to which a virtual environment submerges the perceptual system of the user" (Biocca and Delaney, 1995). When a system constructs an illusion of an alternative reality, the more comprehensive and believable the sensory information it delivers is, the more immersive the system is considered to be (Slater et al., 2009). A VR system can be evaluated according to the degree of *immersion* it provides (through technology features, such as stereoscopy, image quality, etc.). For example, a 2D video game is considered non-immersive, whereas VR delivered via an HMD is considered fully immersive. It has been suggested that the greater the level of *immersion*, the stronger the sense of *presence* (Cummings and Bailenson, 2016; IJsselsteijn et al., 2001; North and North, 2016; Slater and Wilbur, 1997), indicating that in order to create a powerful effect of *presence*, the use of *immersion* is essential.

Two platforms for two types of presence

The two types of technological platforms we briefly reviewed here – the SAR and the immersive VR – use different means to create engagement and motivation: a SAR uses the nature of human beings as social entities who are driven by social interactions (Jung and Lee, 2004). Being embodied, the SARs make use of the significance of the physical presence of a social partner (Kwak et al., 2013). VR uses immersion and the spatial presence effect for creating engaging activities by submerging users in a virtual world. Both SARs and immersive VR are interactive interfaces that provide automated engaging environments, which can be used for clinical purposes. These technologies rely on cognitive and psychological principles of perception and information processing that convert sensory cues into a holistic sense of *presence* (Jung and Lee, 2004). Moreover, these technologies allow for simulations of social and spatial experiences, which, in turn, create a sense of "unmediated experiences" (Lee, 2004). These are intuitive and simple user interfaces which are advantageous, especially when designing applications that are targeted at the older population.

Which of the two types of presence – social or spatial – is more powerful in creating an engaging and motivating experience? Considering the great effort and resources invested in finding engaging tools that would lead to adherence to therapeutic programs, a clear comparison between these platforms is necessary.

Physical embodiment vs. a virtual experience

The difference between interacting with a physically present social robot versus engaging in virtual experiences has been explored (Bainbridge et al., 2011; Brooks et al., 2012; Kiesler et al., 2008; Lee et al., 2006; Li, 2015; Nishio et al., 2021; Wang and Rau, 2019). The studied virtual agents ranged from telepresent robots – i.e., robots that do not share the same physical space with the user (Bainbridge et al., 2011) – to simulated virtual agents that express the same social cues as a physically embodied robot (e.g. Brooks et al., 2012; Kiesler et al., 2008; Lee et al., 2006). These studies concluded that physical presence and the robot's embodiment play a crucial role in social interaction when people interact with an automated social entity; in other words, they found that social robots are preferred over virtual environments. However, these studies compared an interaction with a physical entity to an interaction with a virtualized form of that entity [e.g., a virtual simulation of an existing robot (Wang and Rau, 2019)]; thus, they did not take full advantage of the sense of spatial presence that immersive virtual reality can offer nowadays, and the results may therefore have been affected by the tendency to favor interaction with a co-located entity over interaction with a virtual one (Biocca and Harms, 2002). We posit that in order to achieve high engagement rates by users, it would be advantageous, when comparing a SAR with a virtual environment, to capitalize on what each platform does best: SARs have the potential to create a sense of *social presence*, and immersive virtual reality has the potential to create a sense of *spatial presence* (see Fig. 1). It is important to note that this distinction is not binary—there exists the potential to create virtual characters in VR that establish a sense of social presence, and spatial presence can be achieved by using augmented reality with physical robots.

A question that thus remains yet unanswered is what would users prefer—a physical embodiment or virtual experience— when the virtual setting does not attempt to replicate a social situation, but rather offers an immersive experience that submerges them in the virtual surroundings?

Using social robots or VR for cognitive training is a relatively recent endeavor. Therefore, new forms of cognitive training delivered by these platforms can enrich and contribute to the research and development of cognitive training. Testing the experience of the target population of the training, i.e., older adults, versus a younger population can refine the source of differences in perception of the two platforms.

Here, we implemented established cognitive training tasks on two technological platforms, one which is able to create a sense of *social presence* – a socially assistive robot – and one which is able to create a sense of *spatial presence* – immersive virtual reality. Our goals were: (1) to design two user experiences – based on either social or spatial presence – for the purpose of cognitive training; (2) to evaluate how young and old adults perceive the experience with the two platforms; (3) to test how young and old adults evaluate training with the two platforms, and their motivation to continue training with either one; and (4) whether interpersonal differences such as age, gender, personality traits, and empathic tendencies correlate with their preference for one platform over the other.

2. Methods

2.1. Design

We designed and tested two technological platforms for administration of cognitive training to users. We endeavored to build a system that would leverage each platform's unique advantage: a social robot and immersive VR. The design relies on three main principles: modularity, presence, and gamification, as detailed below.

Modularity

Our objective was to build a modular system, into which blocks of cognitive training can be added, as well as removed. Underlying the system's structure is the following repeating three-step sequence: a cognitive task, a reward, and a presence-enhancing experience during a break. In order to keep the users engaged, each sequence of task-reward-presence-enhancing-experience was designed to be different from all of the other sequences. Furthermore, the sequences were designed to best leverage the unique advantages of each of the two platforms. For example, to create the sense of a versatile social entity, during one of the breaks, the robot was relaxed, and it meditated, and in another one it danced energetically. In the VR setup, the players were sailing on the ocean during one break and were riding a motorcycle during another (see video S1, S2). After a task completion, the reward in the robot interaction could be either an encouraging and motivating response, or alternatively, a comforting and empathic one, in cases when less than half of the responses were correct. On the VR platform, where users went on a journey (details below), the reward was the removal of an obstacle on the path of the player. For example, in one of the scenes, the user was diving underwater; a school of fish blocked the player's ability to progress underwater. Upon completion of a cognitive task, the school of fish swims away, clearing the path; or, in another scene, a box with the keys for the next task is unlocked once the cognitive task is complete.

A common problem among cognitive-training studies is that their effectiveness is difficult to prove – gamified training methods are very different from one another, and they are often tailored to a single specific context (Kueider et al., 2012; Shah et al., 2017). Therefore, it is hard to replicate the tasks in different settings or compare the results to previous studies (Shah et al., 2017). Our aim was to demonstrate the use of well-tested traditional tasks (Au et al., 2015; Koch et al., 2018), implemented in a gamified setting using two different technological platforms.

We embedded the cognitive task as an integral part of the platform: In the immersive VR, stimuli were presented in a way that fit the context of that setting (e.g., on the pages of a book placed in front of the user), and in the interaction with the robot, the stimuli were placed on display components (LED arrays) that are an integral part of the robot (see Fig. 3).

Cognitive tasks

The cognitive training, both with the social robot and with the VR, was comprised of two different cognitive tasks – N-back and Task-Switching – commonly used in cognitive training (e.g., (Basak et al., 2008; Jaeggi et al., 2011; Karbach and Kray, 2009)).

N-Back task. Task Description: The *N-Back* task aims to improve the ability to update working memory (Au et al., 2015). It comprises a series of stimuli of different characteristics (e.g., shapes, colors, numbers, etc.), presented one after the other. Participants are required to notice when a certain stimulus is identical to the one that appeared n steps before it and press a response key as soon as they do. In the simplest implementation, $N = 1$, and this value increases with the increase in training difficulty.

Custom Implementation: In our implementation, the task is composed of colored shapes that can appear either on the right or on the left side of a display, one shape at a time. The task is to press a response key when the displayed shape is the same as the one shown N steps before.

1 *Basic difficulty level:* At the lowest difficulty level, the task is to press a response key when the displayed shape is the same as the one shown one step before ($N = 1$).
2 *Intermediate and advanced difficulty levels:* Key press is required when the displayed shape is the same as the one shown two steps before it ($N = 2$). At the advanced level, the number of 2-back occurrences increases.

Task-Switching task. Task Description: Task Switching is another example of a task that involves executive functions, aimed to improve the ability to shift attention from one task to another (Koch et al., 2018). This task can be implemented in many forms (e.g., (Rubin and Meiran, 2005; Strobach et al., 2012; Wilckens et al., 2017)), which all share the same principle of rapidly switching attention from one feature of a stimulus (e.g., the shape) to another (e.g., the color) and responding appropriately. For example, a stimulus may be presented on the top or bottom of a computer display, and the participant must pay attention either to the shape of stimulus or to its color, based on the vertical location of the stimulus (Green et al., 2012).

Custom Implementation: In our implementation, the **Task-Switching** series is composed of shapes in four different colors, which can appear on the upper or lower part of a display. The task requires that participants respond only to the color of the stimulus if it appears in the upper part of the display, and only to the shape of the stimulus if it appears in the lower part of the display.

1 *Basic difficulty level*: At the lowest difficulty level, a single shape is displayed on the right or left side, either at the lower or at the upper part of the display. Participants are asked to press a response key if a specific pre-defined color (e.g., green) appears on the upper part of the display, or if a specific shape (e.g., square) appears on the lower part of the display, while ignoring the irrelevant property (shape or color, respectively).
2 *Intermediate and advanced difficulty levels:* The series of stimuli is composed of pairs of stimuli that can appear in the upper or lower part of the display. Participants are asked to press a key if the color of both stimuli in the pair (appearing in the upper part) or the shape of both stimuli (in the lower part) is identical (see Fig. 2). At the advanced level, the congruency rate decreases; that is, there are more cases where the correct response of the relevant task, according to stimuli appearance (press the key or not) is opposite to the correct response for the irrelevant task. For example, if a stimulus appears on the upper part of the display and both stimuli have the same color, but the shapes are different, the correct response for the relevant task, i.e., color comparison, is to press the key, while the correct

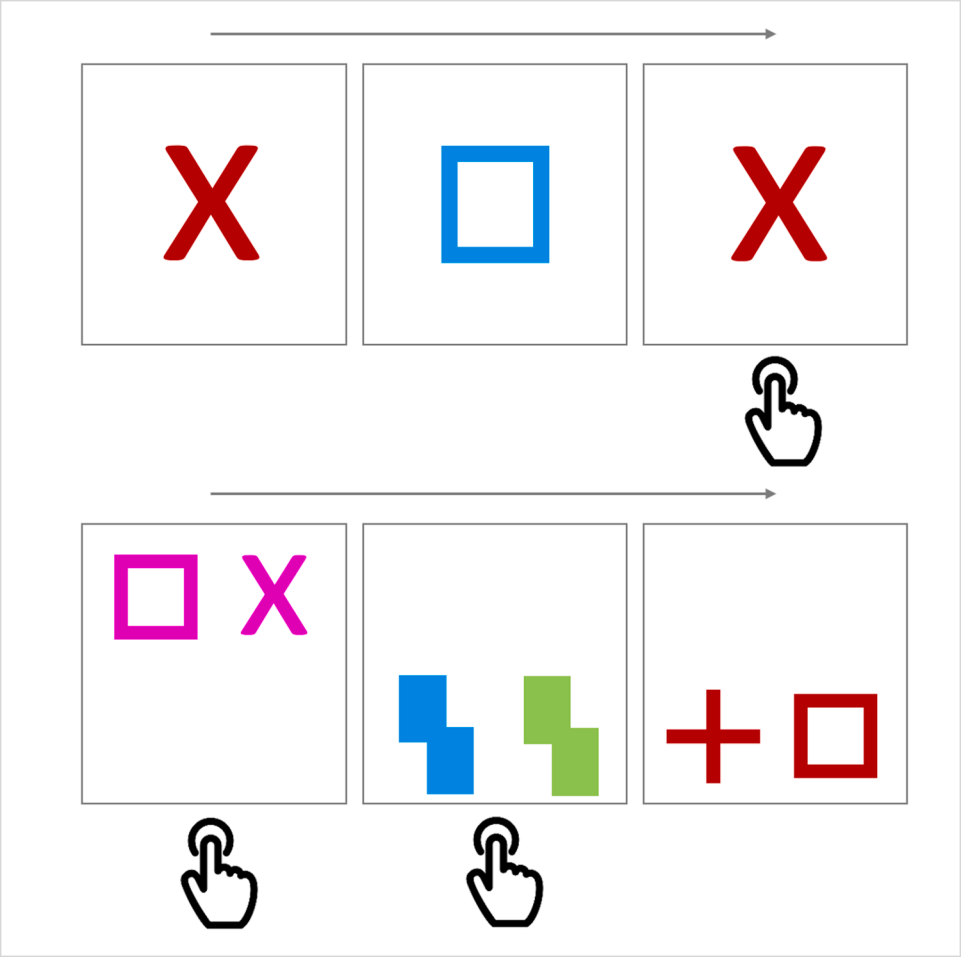


Fig. 2. An example of the stimuli series at the advanced level of each of the two tasks. Top: N-Back task: Participants are asked to press a response key when they see a stimulus that is identical to the one that appeared n steps beforehand. At the advanced level, $n = 2$, as demonstrated here: an X is followed by a square, and another X. After the third stimulus appears (X), the required action is to press the response key, since it is identical to the first stimulus, which appeared two steps before it. Bottom: Task Switching: Participants are asked to press a response key if a pair of stimuli appear on the top of the screen and both have the same color, or if a pair of stimuli appear on the bottom of the screen and both have the same shape. In the sequence shown here, the response key should be pressed when the first pair of stimuli appears (the color of the shapes is identical, and the pair appears on the top of the screen), and when the second pair of stimuli appears (the shape of the stimuli is identical, and the pair appears on the bottom of the screen). The response key should not be pressed after the third pair of stimuli appears (the color of the stimuli is identical, and the pair appears on the bottom of the screen).

response of the irrelevant task, i.e., shape comparison, is to not press the key. Additionally, the number of switching steps increases, i.e., changes in the type of the task between subsequent stimuli are more frequent.

Presence

A fundamental component of our design was creating a sense of *presence*. We created a training framework in which the environment is similar to a real-life situation in order to enhance engagement and excitement during training. We intended to generate a sense of *spatial* or *social presence,* our aim being to increase involvement and motivation of the users. We used two technological platforms that can effectively produce these effects (e.g., Jung and Lee, 2004; Lee et al., 2006; North and North, 2016; Slater and Sanchez-Vives, 2016). Our approach in producing a sense of presence was to utilize the unique strengths of each platform by providing presence-enhancing experiences during the breaks between tasks and when delivering feedback: to create these experiences with the robot, we used social responses and the robot's human-like behavior (e.g., expressing anthropomorphic gestures and socially aware responses throughout the training); to create these experiences in the immersive VR, we created highly detailed real-life scenes with a variety of events (the specific implementation is detailed below). By creating a social experience with a human-like entity, we aimed to evoke emotions that social connection can elicit such as empathy, amusement, encouragement, and solidarity. By providing a life-like spatial environment, we aimed to arouse emotions that different surroundings can evoke such as excitement, thrill, and curiosity.

Gamification

A main challenge in cognitive training methods is to keep users engaged while they perform multiple repetitions of the tasks (Park and Bischof, 2013). It is therefore essential to develop tools to reduce tedium during cognitive training (Park and Bischof, 2013). Gamification of cognitive training has been found to increase engagement and boost participants' motivation (Lumsden et al., 2016; North and North, 2016), amplifying arousal, enjoyment, and challenge. *Rewards* and positive reinforcements are important components of gamification (Skinner, 1990). An additional important element of gamification is a *storyline* (Sailer et al., 2014): a meaningful narrative that frames the activities, thus eliciting interest and positive emotions and contributing to the feeling of "flow" - a subjective state where one is "completely involved in something to the point of forgetting time, fatigue, and everything else but the activity itself" (Nakamura and Csikszentmihalyi, 2014).

We applied the principles of gamification to each of the two platforms by introducing a *storyline* and *rewarding* mechanism in each: As part of the storyline, the robot was portrayed as a training coach, who goes through the training with the participant, and encourages them to perform the cognitive tasks, interspersed with activity breaks; the activity breaks included a stretching session, an invitation to dance in place, to meditate, etc. At the beginning of the interaction, the social robot introduced itself as a coach, who will do the training alongside the participant. Before the training started, it performed warm-up exercises, and prepared for the training; following each training phase, the robot rewarded the users for their progress, either by expressing an empathic response in case of poor performance, or by complimenting the user for their success. In between training sets, the robot performed an action in response to the previous set (e.g., meditating after a difficult level), and

commented on the upcoming set ("Let's diversify the tasks"; "This is fun"; "It's exhausting"; "We're almost done"; "This is the final step", etc.). The robot expressed different moods and levels of energy as the training progressed: energized and fresh at the start of the session, stressed when the training got more difficult, relaxed after meditating, bored after repeating sets of the same task, happy when the training was almost complete, and joyful at the end of the training, using enthusiastic gestures and dancing victoriously.

The VR scenes composed a storyline of a journey in an unfamiliar area, in which the cognitive tasks were presented in the context of the current settings of each scene. The VR game introduced six scenes, each containing an obstacle that prevented the storyline from progressing. Upon completion of the tasks, the obstacles were removed; the removal of the obstacles constituted the reward.

2.2. Implementation

2.2.1. Training with the social robot

Hardware. We used a small humanoid robot (JD humanoid, EZ-ROBOT, Canada; 22.9 × 15.2 × 1.8 cm in size, with 16 degrees of freedom) for this experiment. The robot had four 3 × 3 RGB LED blocks on its body: two blocks located where its eyes are, and two in the chest area (see Fig. 3). The visual stimuli of the N-Back task were presented on the chest screens, and the stimuli for the Task-Switching task were presented both on the eye screens (upper display area) as well as on the chest screens (lower display area). We used an external speaker (Sony SRS-XB21) to maintain high sound quality, and a wireless keyboard was used as the input device when the participants performed the cognitive tasks.

Software. The interaction protocol was implemented using the C# programming language and was connected to the robot by the EZ-Robot SDK.

Interaction. Our aim was to give the robot social characteristics that elicit in the players a sense that the robot is aware of them, is aware of the progression of the training, and that it has empathy for them. To that end, we:

1 programmed the robot to address the player directly with the appropriately gendered pronoun (in Hebrew, there is a masculine and a feminine form for the second-person pronoun 'you'), according to the gender that the experimenter indicated during setup.
2 avoided repetitions of the same response during an interaction with a single player; the responses were drawn from a predefined set of diverse encouraging and comforting responses.
3 programmed the robot to respond empathetically to errors made by the player (e.g., "It's hard for everyone, you'll get better with more practice"). These responses were generated automatically when the success rate on the last set of 20 trials was lower than 50%.
4 programmed the robot to encourage the player after they performed well (e.g., "You are getting better," "You are a pro!"). These responses were generated automatically when the success rate on the last set of 20 trials was higher than 70%.

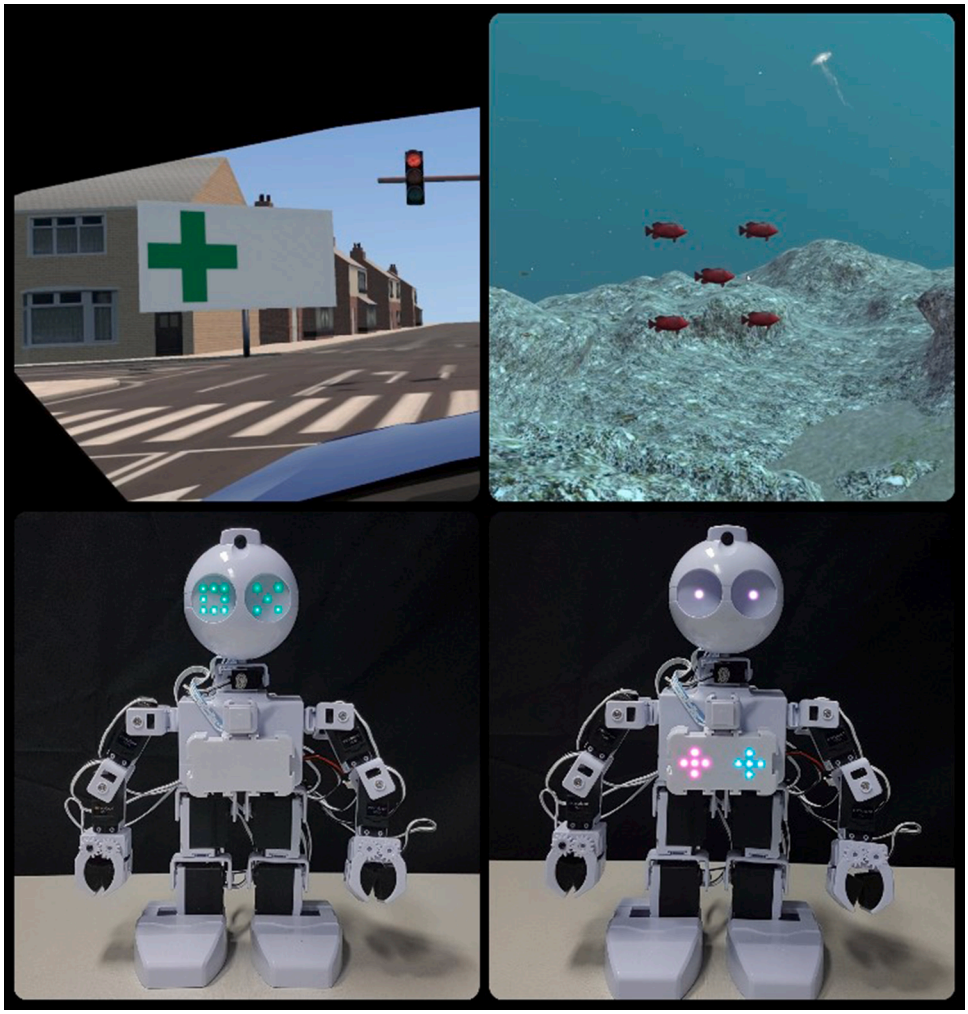


Fig. 3. Examples of stimuli-display implementations. Top panel: we embedded the stimuli for the cognitive tasks in the VR environment either by projecting them on objects such as a billboard in an urban surrounding (top-left panel) or by re-arranging objects in order to create a stimulus, such as several fish in the ocean, which grouped together to create a single stimulus (top-right panel). Bottom panel: The stimuli appeared on four 3 × 3 RGB LED screens: two were located in the robot's eyes (bottom left panel), and two on its chest area (bottom right panel).

5 positioned the robot to face the player; whenever the robot ended up in a position that was not facing the player (e.g., after dancing), it was pre-programmed to move back to a position facing the participant.
6 programmed the robot to react to the levels of difficulty of the task (e.g., "This is so confusing," "This was a hard one, right?") and make comments when the task type was changing (e.g., "I think should spice it up," "Ready for a challenge?").

The robot was further programmed to have a sense of humor, to be cheerful and playful, and to have different (positive or neutral) moods, such as happiness, calmness, and amusement. It expressed human-like seemingly spontaneous body language expressions such as scratching its head, stretching, waving, sitting, whistling, and seeming to look randomly around the room. Throughout the presence-enhancing experiences during breaks, the robot was programmed to perform different activities, in order to present a variety of emotional states and actions. Each player was exposed to the full set of these emotional states and activities, which were presented via the following:

1 The robot performed warm-up exercises before the training started.
2 It performed a brief physical workout, which included pushups and crunches, to signal that it is energized and ready to engage in the training session with the participant.
3 It meditated and performed some yoga exercises to relax from the stress of the tasks.
4 It whistled when it was bored, yawned, and looked around to relieve boredom.
5 It stretched to freshen up.
6 It danced happily when the training was about to be completed (see video S1).
7 It performed a celebratory gesture, followed by a bow when the training was finished.

2.2.2. Training with immersive virtual reality

Hardware. We used an immersive VR headset (VIVE Pro, HTC, Taiwan, and Valve Corporation, Washington, USA) and a standard wireless keyboard.

Software. The game was implemented using the Unity software, version 2018.2.5f1 and the C# programming language. Graphical objects and packages were imported from the Unity Asset Store.

Interaction. The game was carefully designed to create a spatial presence effect. The scenes included high-quality realistic graphics and sounds, and a real-life-like movement of objects and characters. To maintain interest throughout the game, it was comprised of high-detail dynamic environments, which included changing weather, moving objects, and simulated movement of the user through space (see Fig. 5). Recognizing that different people may relate to different scenarios in the VR, we implemented six different scenes to increase the potential of the VR setup to appeal to a wide range of players. To keep the game progressing as naturally as possible, breaks between task phases were embedded as part of the storyline. That is, during the breaks there was a progression in the storyline. For example, during one of the breaks, the virtual car that the player was driving was cruising down a city street, the radio was on, and pedestrians were walking by, chatting with each other (see video S2).

Donning the VR helmet, the players entered the following scenario: they arrived in a rented apartment, as part of a journey to an unfamiliar area, and were given instructions on how to progress through different landscapes, on land, over- and under-water, and in the air. The game had six different scenes, each with an obstacle preventing the journey from continuing. To remove the obstacle, the player had to perform a cognitive task. At each of the six scenes, corresponding to the six experimental phases (outlined in Fig 6), a series of shapes appeared on objects in that scene in a way that fit the context of that setting. For example, when two ships blocked the progression of a player's boat in the sea, the stimuli appeared on the flags of one of the ships (see Fig. 5D). Once the player completed the task, the obstacle was removed, and the journey continued to the next scene (see video S2). The final goal was to complete the journey and make it back home. Therefore, the player needed to remove the obstacles as quickly as possible. The six scenes are detailed below:

1) A Rental Apartment: The game begins in the living room of a rental apartment, where a phone call is received from a tour guide telling the player about the journey. The player sits next to a desk. On the desk is a book and a locked box with car keys inside it, which the player needs to get to unlock and start the car. To open the box and get the car keys, the player must perform an N-back task. The shape series is presented on the book pages (see Fig. 5A). When the player completes the task, the box opens, and the keys are revealed.
2) Urban Driving: The player drives a car in a busy urban environment. Different vehicles are driving around, people are passing by, and country music is playing on the radio. At one junction, the traffic light turns red. For the light to turn green again, the player must perform an N-back task. The stimuli series is presented on a billboard in front of the car (see Fig. 5B). When the player finishes the task, the traffic light turns green, and the ride continues.
3) Underwater: The player is diving in the ocean, passing different fish, sea plants, and marine rocks. After a while, a school of fish blocks the way, so the player has to complete an N-back task to continue diving. This time, the series of shapes is created by fish grouping together into the possible shapes while changing their color (see Fig. 5C). When the player completes the task, the fish clear the way and the player can swim forward.
4) Sailing: The player is sailing in a boat on the ocean when suddenly two ships block the boat's path and it is no longer possible to continue. In order to make the ships move aside, the player must complete a Task-Switching task. The series of shapes is now presented on four flags hanging on one ship – two in an upper part of the ship and the other two at a lower part (see Fig. 5D). When the player completes the task, the ships sail away, and the player sails ahead.
5) Off-road riding: The player rides a motorcycle in a mountainous area until encountering a closed gate. The stimuli of the Task-Switching task are presented on two traffic signs next to the gate (see Fig. 5E). Once the task is completed, the gate opens, and the driving continues.
6) Flight: The player flies an airplane on the way home. In order to land the plane, the player must perform a Task-Switching task, which is presented on the dashboard in the cockpit (see Fig. 5F). When the task is completed, the landing begins.

2.3. Experimental protocol

Participants

Sixty-four adults participated in the experiment - 32 older adults (59–82 years old, 69.6 ± 5.7) and 32 young adults (21–35 years old, 25.8 ± 3.2). Each age group was composed of 16 males and 16 females. All the participants were healthy, without any neurological abnormalities. Participants were recruited through standard procedures, such as ads on campus, social media, and mailing lists. They received monetary compensation for their participation.

Protocol overview

Written informed consent was obtained from all the participants upon arrival to the laboratory. Written informed consent for publication of identifying images in an online open-access publication was obtained from the persons photographed. The experiment was approved by the

institutional review board of the Ben-Gurion University. All experimental procedures were performed in accordance with this ethical approval. The experimenter instructed the participants how to perform the cognitive tasks and asked them to fill out a personality questionnaire – the Big Five Inventory (John et al., 1991), using its validated Hebrew version (Etzion, 1998), and an empathy assessment - Interpersonal Reactivity Index (Davis, 1983), using its validated Hebrew version (Even, 1992). Older participants underwent a cognitive evaluation using the Mini-Mental State Exam (MMSE) (Folstein et al., 1975). Young participants did not undergo the cognitive evaluation since, as university students, they had passed entrance exams, which indicated they had preserved cognitive functioning.

They were then semi-randomly assigned to start the cognitive training with either a social robot or an immersive VR system (see details below), such that an equal number of young and older adults from either gender started training with the VR as those who started training with the robot. Once they completed the cognitive training with one system (VR or robot), they had a 3-minute break, after which they performed the training with the other system (robot or VR, respectively; see Fig. 6). After they completed both training sets, they filled out a custom-made questionnaire (see below).

Each task was comprised of three phases, in each of which a series of 20 stimuli were presented (for a total of 60 stimuli per task). Each stimulus was presented for two seconds, during which participants were required to respond, by pressing or not pressing a key. There was a one-second interval between consecutive stimuli. Each visual stimulus was composed of either one or two shapes, out of four possible shapes (see Fig. 2). Each task had three possible difficulty levels: baseline, intermediate, and advanced. Participants automatically advanced up the difficulty levels if they responded correctly on at least 75% of the stimuli in the previous phase (that is, on at least 15 stimuli presentations out of 20). To keep the degree of difficulty uniform across participants at the same level, there was a constant number of required key presses at each difficulty level; similarly, there was a constant number of switching steps, and a constant congruency rate per level.

Participants were instructed to press the space bar of a standard computer keyboard as fast as possible when the visual stimuli that were displayed to them had certain characteristics, as explained above. At the end of each of the three phases, participants were told – by the VR system or by the robot – the number of correct responses they had made out of the twenty possible correct responses. They then received a one-minute break, during which they took part in a presence-enhancing experience, as detailed above.

To maintain uniformity of the tasks between the two experimental platforms (the robot and the VR), the tasks were designed to have the same duration (approximately 23 min), and the timing of the presence-enhancing experiences during breaks between the tasks was similar as well. Furthermore, game progression and difficulty levels were implemented identically across platforms, the stimuli were exactly the same (same shapes, same colors), and in both environments the exact same pre-recorded instructions were used. With both platforms, participants got rewards (social feedback by the robot, or game progression in the VR) for completing each phase and experienced diverse system behaviors throughout breaks.

Questionnaire

At the end of the experiment participants were requested to answer a questionnaire that included both open-ended and Likert scale questions (on a scale from 1 to 7). The questionnaire included questions about their experience with the robot and with the VR, such as the degree of enjoyment, interest, excitement, level of difficulty, level of comfort, level of robot sociability, and the preferred system for future training. The questions about future use were deliberately ambiguous (e.g., "use for a few hours") to enable participants to answer about an order of magnitude of training rather than exact time duration. To evaluate the participants' perception of social and spatial presence when using the two technological platforms, we asked them about these directly. The questions asking about the extent to which they experienced *spatial presence* were taken from commonly used presence questionnaires (Slater et al., 1994; Witmer and Singer, 1998), and participants reported their feeling of *social presence* by answering questions about contributors to a social-presence effect in human-robot interaction – sense of empathy (Leite et al., 2013), cooperation (Biocca and Harms, 2002; Oh et al., 2018), and anthropomorphism (Fink, 2012).

Hypotheses

We hypothesized that: (1) users would have a preference for the social-robot platform over the VR platform, based on previous literature (Bainbridge et al., 2011; Brooks et al., 2012; Kiesler et al., 2008; Lee et al., 2006; Li, 2015; Wang and Rau, 2019); (2) users would maintain their preference for long-term training; and (3) interpersonal differences, such as age, gender, personality traits, and empathic tendencies, would correlate with this preference.

3. Results

Statistical analysis

To make sure that there were no differences in performance between the two platforms, which could potentially cause a bias in preferences, we calculated and compared the success rates of participants on each platform. The success rate on a task was calculated as the proportion of correct responses out of the possible 20 correct responses in each phase. The assumption that performance data came from a normal distribution was not confirmed ($p<0.001$, Shapiro-Wilk Normality Test), and therefore we used the Kruskal Wallis to compare results between the two age groups, and the Wilcoxon signed-rank test to perform a paired comparison of the success rates between the two platforms. We calculated the Cronbach's alpha coefficient to measure the reliability of the custom-built questionnaire, and found it to be 0.695. To analyze group differences in preferences and perception we used the non-parametric Chi-square test, and we used Fisher's exact test to measure associations between variables.

Performance

Performance did not differ significantly between the two platforms on either the N-back task ($p = 0.845$) or the Task-Switching task ($p = 0.287$; Wilcoxon signed-rank test). There was no trend for choosing one of the two platforms as more difficult than the other ($\chi^2=2.77$, df=1, $p = 0.096$). The success rate of the older users was significantly lower (N-back: 0.82±0.1, Task-Switching:0.90±0.1) than that of the younger ones (N-back: 0.94±0.07, Task-Switching:0.97±0.04) across both types of tasks and both technology platforms ($p<0.001$, Kruskal-Wallis Test). The average MMSE score of the older adults was 28.2 ± 1.7.

We designed the cognitive training sets to have the same duration on both platforms. Participants were given the option to hear the instructions replayed, which led to small differences in the duration of the sets across participants. On average, old participants took longer to complete the tasks on both platforms, and the interaction with the robot was slightly shorter both for older and for younger participants (VR: 26.9 ± 2.7 min (old), 25.1 ± 3.57 (young); Robot: 25.9 ± 2.3 min (old), 24.4 ± 2.0 min (young)). Most people had no prior experience with VR (64%) or with social robots (70%). The order in which the participants trained on the two platforms (whether users first trained with the VR or with the robot) did not affect the users' preferences when they were asked which platform they liked better ($p = 0.089$, Fisher's exact).

Perceived experience

To examine whether participants perceived the robot as a social

entity, they were asked to rate various aspects of the robot's social characteristics (whether they perceived it to have human-like characteristics, to be cooperative and empathetic). The results demonstrate that participants indeed considered the robot as a social entity (see Fig. 7). Human characteristics were attributed to the robot by 63% of the participants (χ^2=4.26, df=1, $p = 0.039$), 74% reported that they felt that the robot had cooperated with them (χ^2=11.79, df=1, $p = 0.001$), and 70% felt the robot was empathic (χ^2=8.96, df=1, $p = 0.003$).

To understand the extent to which the participants felt present in the VR experience, they were asked about their sense of presence and their awareness of the world outside the virtual reality. The results demonstrate that the participants indeed felt present during the experience. The overwhelming majority of participants – 97% – reported having felt a sense of presence (χ^2=52.26, df=1, $p<0.001$), and 66% were not aware of the outside world (χ^2=5.255, df=1, $p = 0.022$). Seventy six percent felt the experience was realistic (χ^2=14.75, df=1, $p<0.001$). When asked whether the experience was similar to watching a regular screen display, 87% replied negatively (χ^2=18.64, df=1, $p<0.001$).

When asked to compare their experience between the two platforms, 64% of participants found the VR experience to be more exciting (χ^2=4.89, df=1, $p = 0.027$), 69% reported they enjoyed it more (χ^2=7.36, df=1, $p = 0.007$), and 71% found it to be more interesting (χ^2=9.00, df=1, $p = 0.003$) than the interaction with the robot. At the same time, 65% of the participants reported feeling more comfortable with the social robot than with the VR (χ^2=4.59, df=1, $p = 0.032$). There was no significant difference between the reported sense of control between the two platforms, with 60% of the users reporting feeling more in control in the interaction with the robot (χ^2=1.97, df=1, $p = 0.160$). In accordance with the results showing similar performance rates between the two platforms, 50% of the participants reported it was easier for them to concentrate when training with the VR, and the other 50% reported they concentrated better when training with the robot.

Additionally, we tested whether aspects of presence – either *social* or *spatial* – such as perception of the robot as empathic, or unawareness of the outside reality when in the VR, are correlated with finding the platform more interesting, exciting, and easy to concentrate on, or reporting a greater sense of control when using that platform. Those who considered the VR experience to be realistic, also concentrated better when using the VR platform (68%), while those who did not consider the VR experience to be realistic felt they were better able to concentrate with the social robot (83%, $p = 0.005$, Fisher's exact). Participants who found it easier to concentrate in the VR environment had a larger ($p = 0.002$) tendency to prefer the VR platform in the short term (86%), while those who reported that they concentrated better when using the social robot, did not show a clear preference for either of the platforms over the other in the short term.

When asked which of the two platforms they preferred to train with, the majority of participants (66%) preferred to use the VR, compared to 34% who preferred to interact with the social robot (χ^2=5.78, df=1, $p = 0.016$). We thus refuted hypothesis 1. The preference significantly correlated with how interesting, exciting, or enjoyable they found the platform of choice ($p<0.001$, Fisher's exact). However, this preference for the VR was most pronounced in the short term, and diminished in the long term. That is, when asked which platform they would like to train with over the course of hours, the difference between preferences for the platforms shrunk to just 54% percent choosing VR over 46% who preferred the robot (χ^2=0.32, df=1, $p = 0.572$); when asked which platform they would like to train with over the course of days, participants' preferences were split exactly in half between the VR and the social robot (see Fig. 8). We thus refuted hypothesis 2. These preferences in the short and in the long term did not correlate with age ($p \geq 0.779$, Fisher's exact).

Interpersonal differences

We further examined whether age had an effect on user preferences and perceived experience. Age did not correlate with preference ($p = 0.779$, Fisher's exact), and on both platforms, the perceived experience of young and older adults was similar in all aspects (see Fig. 7); Neither *social presence* perception ($p = 0.544$, 0.580, 1, for cooperation, empathy, anthropomorphism, respectively) nor *spatial presence* perception ($p = 0.494$,0.781,0.358, for presence, actual reality unawareness, realistic feeling, respectively) correlated with age. Yet, significantly more older adults than younger adults reported feeling excited by the VR system more than by the social robot ($p = 0.034$, Fisher's exact).

Top: Users' perception of the robot's social presence. *Bottom:* Users' perception of a spatial presence within the immersive VR environment.

On the Big Five Inventory questionnaire, participants rated themselves on a scale of 1–5 in response to 44 statements representing five main personality traits: Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness-to-experience (John et al., 1991). On the IRI questionnaire, they similarly rated themselves on a scale of 1–5 in response to 28 statements representing four empathy scales: Empathic concern, Perspective taking, Personal distress, and Fantasy (Davis, 1983). We calculated the median value of the score for each personality trait and empathy tendency, and used it to group participants into a "high" and a "low" group per trait and empathic tendency. Those with values above the median were in the "high" group, and those with values equal to or below the median were in the "low" group. Neither personality dimensions ($p = 0.054$, 0.561, 1, 0.394, 1, for Extraversion, Agreeableness, Neuroticism, Conscientiousness and Openness-to-experience dimension, respectively) nor empathic tendencies ($p = 0.573$, 0.578, 0.257,0.578, for Perspective Taking, Personal Distress, Empathic Concern, and Fantasy scale, respectively) correlated with preferences. Gender did not correlate with participants' preference either ($p = 1$). We thus refuted hypothesis 3.

Qualitative analysis

Unprompted, the majority (80%) of participants expressed their enthusiasm for the experience with the two platforms. They repeatedly commented on the originality of the experience and on how exciting and interesting it was for them. Many referred to the great diversity of environments and sense of presence in the VR system and were highly amused by the interaction with the robot, saying it was very friendly and funny. Surprisingly, the most common complaint (made by 10 participants) concerned the multiple breaks during the interaction with the robot (during which it was dancing, stretching, meditating, etc.), though the number and duration of breaks were identical for both platforms.

It was interesting to note the great diversity in the way participants experienced the two technological platforms. While some participants noted it was hard for them to concentrate in the VR setting, as it contained many stimuli, others found it much easier to concentrate in the VR setting, when they had no distractions from the outside world. Some perceived the VR to be highly realistic, whereas some perceived it as an artificial environment. Some paid greater attention to the robot's body language, whereas others focused on its verbal communication. The differences among participants were also evident from their spontaneous responses during the experiment: some participants joined the robot's activities during the presence-enhancing experiences (e.g., dancing and stretching, see Fig. 4, top-left and bottom-right panels, and video S2), while others simply watched it without actively taking part. In the VR, some participants looked around and explored the virtual environment, while others preferred to look only directly ahead during the entire training session. There were thus anecdotal differences in how users interacted with these platforms, suggesting that future experiments should look at how individual differences not tested here (e.g., education level, digital literacy, attitudes toward technology) may impact the preferences of users, especially in light of previous studies suggesting the importance of personalizing interactions with technology to the specific user's needs and preferences (Eizicovits et al., 2018; Feingold Polak et al., 2018; Kashi and Levy-Tzedek, 2018).

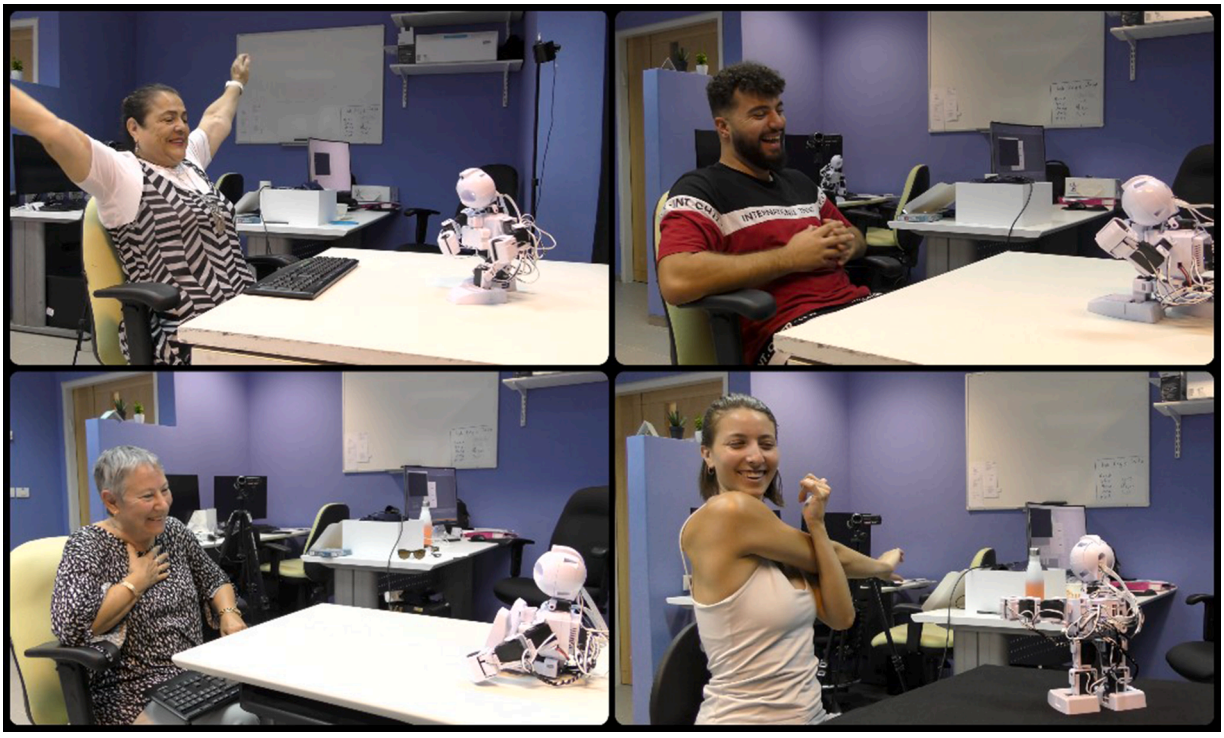


Fig. 4. Presence-enhancing experiences with a social robot. The robot displayed diverse social activities which corresponded to the progression of the training and to the performance of the participants.

4. Discussion

Our goal was to design, implement, and compare two engaging technological systems – a social robot and immersive VR – for the purpose of cognitive training. Unlike previous comparisons between a robot and a VR system, our study capitalized on the unique ability of each platform to create a sense of presence – either social or spatial – thereby comparing the two platforms when each is maximizing its potential to engage users.

This work's main contributions are: (1) the design of two user experiences – based on either social or spatial presence – for the purpose of cognitive training; (2) the evaluation of how users – young and old adults – perceive the experience with the two platforms, rate their training with the two platforms, and rate their motivation to continue training with either one; (3) the correlation between the users' perceived experience and their personal characteristics, such as age, personality traits, and empathic tendencies; and (4) the implementation of an alternative methodology for comparing training with a robot with training using a VR system— a methodology that, rather than effacing the differences between the technologies, leverages each one's potential to create engagement.

In this work, we demonstrated how traditional cognitive tasks can be implemented in gamified environments that are designed to evoke emotional responses and elicit motivation and engagement in the training program, using either social or spatial presence. We created a basic training block, which included the sequence: Task→Reward→Presence-enhancing experience. Using this basic training block, we modified the task presentation, the type of reward, and the specific implementation of the presence-enhancing experience each time it was used. The timing of the three elements of the block remained constant in each presentation of a block, across both platforms. Thus, we created a modular system whereby the same task can be presented with different characteristics each time (e.g., different behavior of the robot, various VR scene styles) between and within platforms, in order to create diversity and interest in cognitive interventions. We tested our implementation by administering the training to young and old adults in a 1-hr session.

We explored the differences between the perceived experience with each one of the platforms, in order to understand the factors that contribute to engagement and to adherence to a training regime. Comparisons between robots and virtual forms of interaction have mostly focused on comparing robots to virtual social entities, which were mostly found to be less engaging than a physically present robot (e.g., Heerink et al., 2008; Jung and Lee, 2004; Leyzberg et al., 2012). The recent advent of immersive VR has enabled the creation of highly engaging experiences in the virtual space through a sense of *spatial presence* (e.g., Dede, 2009; Diemer et al., 2015; Pereira et al., 2014). In parallel, social robots have been developed to create a sense of *social presence* (Breazeal, 2004; Fong et al., 2003). We aimed to identify whether there is a preference for one of these technological platforms – while capitalizing on the unique potential of each one to create a sense of presence, either spatial or social – when used as a method to engage users in a cognitive-training task. We implemented an identical cognitive-training task on two technological platforms. Rather than attempt to make the experience surrounding the task identical (e.g., by creating an avatar of the social robot in the virtual world), we took advantage of the unique benefits that each platform offers – *spatial presence* with the immersive VR system, and *social presence* with the social robot. Thus, we were able to compare how these different types of presence contributed to user motivation.

Preference for the immersive VR

Participants expressed their enthusiasm for both platforms. In contrast to previous studies (e.g., Powers et al., 2007; Shinozawa et al., 2005; Tapus et al., 2009a), participants preferred the virtual experience over interaction with a physically present social robot across age, gender, personality traits, and empathic tendencies. It is likely that the discrepancy between our findings and those of previous studies is a result of the magnitude of the *spatial presence* effect that the virtual environment provided: first, the immersive VR platform, in comparison to other platforms that have been previously compared [e.g., 2D screen



Fig. 5. The scenes in the VR cognitive-training game. Stimuli appeared on objects in a way that fits the context of the scene. The different scenes are: (A) a rental apartment, (B) urban driving, (C) underwater, (D) sailing, (E) off-road riding, and (F) flight.

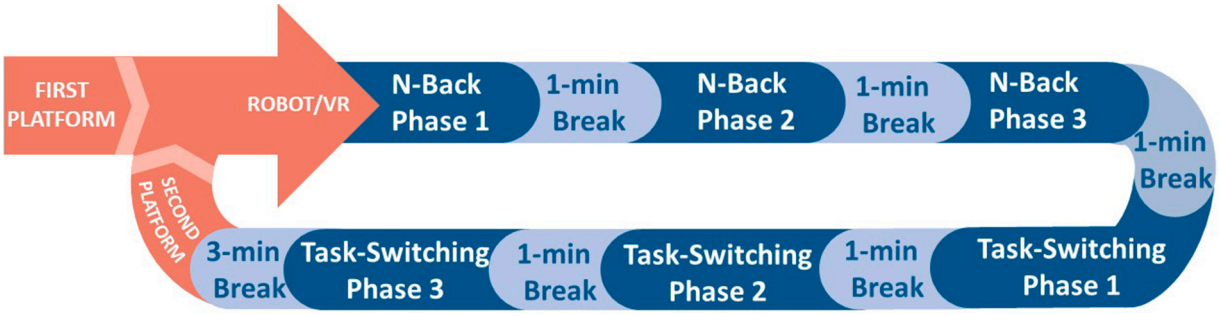


Fig. 6. Experimental flowchart. Participants performed three sets of N-back exercises, followed by three sets of Task-Switching exercises. Half of the participants started with the social robot, and half with the VR. They then repeated these sets of exercises with the other platform (VR or robot, respectively).

(e.g., Bainbridge et al., 2011; Kiesler et al., 2008; Lee et al., 2006; Li, 2015)], is considered to create a stronger *spatial presence* effect (Wirth et al., 2007). Second, in most previous comparisons between VR and social robots, the VR displayed a virtual social agent that would mimic the physical interaction with a robot (e.g., Brooks et al., 2012; Kiesler et al., 2008; Lee et al., 2006; Wang and Rau, 2019)). Our implementation, in contrast, capitalized on the unique strengths that the immersive VR offers, creating a sense of spatial presence (e.g., by rendering highly detailed real-life scenes and using simulated movement through space), rather than attempting to create a virtual *social* interaction.

Participants found the experience with the VR more exciting, interesting, and enjoyable. This is despite the fact that the tasks and the timeline were identical on both platforms, and performance on the task was similar across the platforms. The older adults' performance was poorer than the younger group's, regardless of platform type. These results are in line with previous research showing that older adults find cognitive tasks more challenging than young adults (Bopp and Verhaeghen, 2018; Kray and Lindenberger, 2000; Levy-Tzedek, 2017; Levy-Tzedek et al., 2016). Despite the difference in difficulty between the age groups, the perceived experience was similar for younger and older adults (see Fig. 7); the users in both groups felt *social* and *spatial* presence during the experience.

A sense of presence

Participants reported a sense of presence, realism, and unawareness of the real physical environment when using the immersive VR. When asked about their interaction with the social robot, participants ascribed human characteristics to it and felt it was empathic and cooperative.

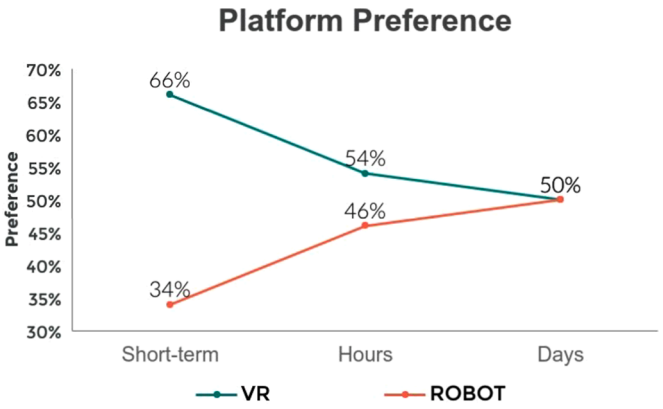


Fig. 8. Preference rates for each platform over different time ranges. Short-term: Participants were asked which of the systems they preferred immediately following the experience with the VR and the robot. Hours/Days: Participants were asked which of the systems they preferred for training sessions that would be administered over several hours or days, respectively. For the short-term experience, almost twice as many people preferred the VR over the social robot (66% vs. 34%), but as the expected duration of the training sessions increased, projected preference rates converged to equal preference for the VR and the robot (50%).

That is, they ascribed a *social presence* to it. The majority of the participants perceived the robot as empathic, cooperative, with human qualities, even though the robot is relatively small-sized, low-cost, and simple in terms of its physical design (e.g., it has no facial expressions). Considering the great effort to build robots with elaborate human-like

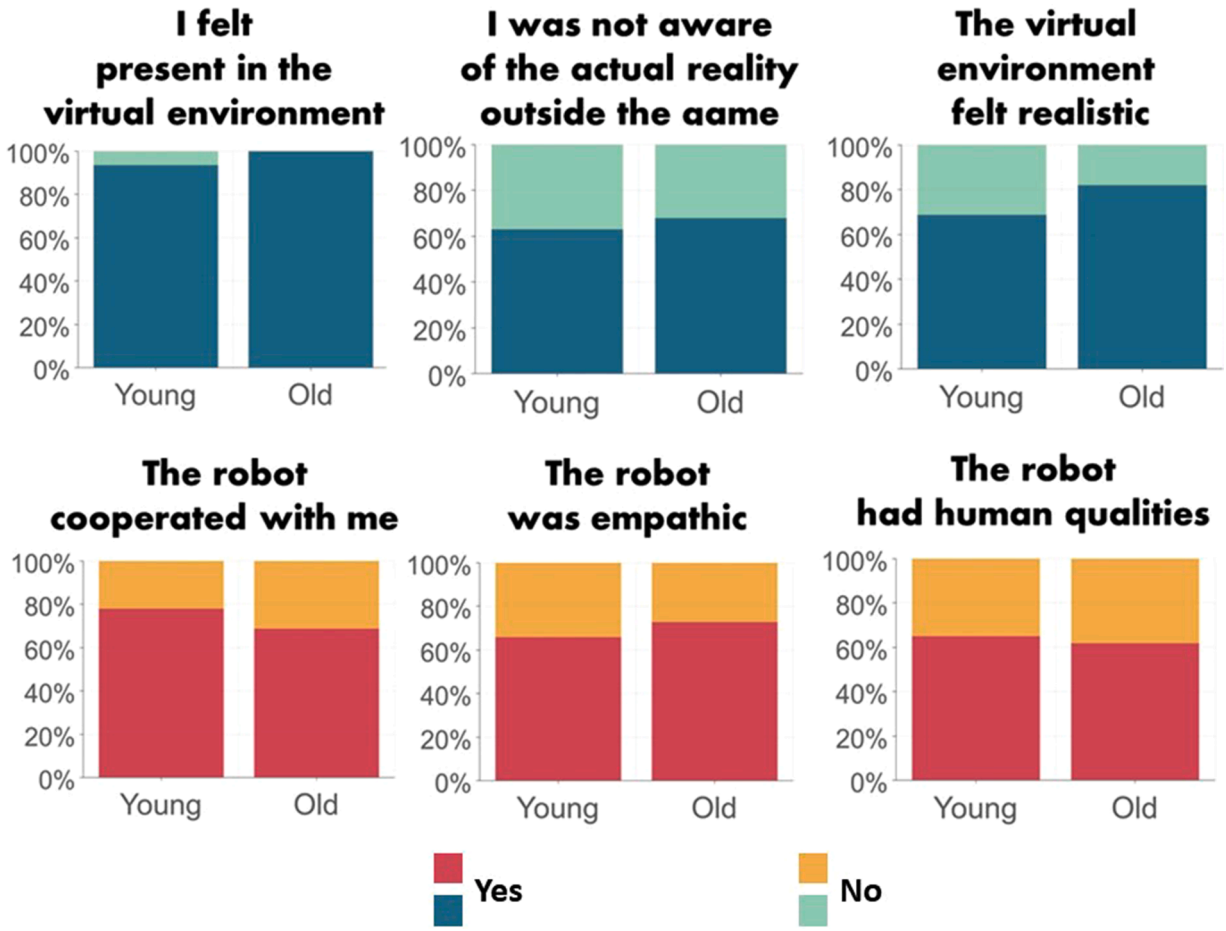


Fig. 7. Younger and older adults' perception of the social and spatial presence when using the two technological platforms.

appearance (e.g., Gouaillier et al., 2008; Pandey and Gelin, 2018; Parmiggiani et al., 2012), the fact that a humanoid with a simple structure can be perceived as social and human-like, and can elicit a variety of social responses from users (see Fig. 4, top-right and bottom-left panels, and video S2) suggests that *functional* resemblance aspects (movements and behavior), rather than a mere accurate *visual* resemblance to a human (e.g., human-size or high-detailed face) underlies sociability and human-likeness in an artificial entity.

Despite the participants' reports that they found the interaction with the social robot to be more comfortable, there was still a clear preference for the VR system when participants were asked which platform they preferred. Common reasons for their preference were the unique feeling of being in another environment ("I felt completely inside it"), the diversity of the environments that kept them interested, and that the emotional experience was more powerful.

It appears that the effect of *spatial presence*, generated by the immersive VR system, was more powerful than the impact of the *social presence* created by the robot. What may underlie this preference is the inherent difference between the immersive VR and the social robot: the VR, in the current implementation, offers an immersive experience, which constitutes a complete storyline taking place at diverse locations. While both platforms offer gamified components, the implementation for VR is similar to common gamified settings such as video games. The interaction with the robot, as implemented here, resembles a conversational setting and offers a diverse set of social behaviors, aimed at establishing a motivating connection between it and the user.

We found that most of the participants who chose the VR platform as the easier one to concentrate on, also felt the experience was realistic, while those who felt they were better able to concentrate with the social robot did not consider the VR setting to be realistic. This result highlights the importance of *presence* in a training framework and is in line with previous findings, which indicate a contribution of realistic settings by VR and social robots to cognitive function improvement, in comparison to traditional and gamified settings only (e.g., Faria et al., 2016; Kim et al., 2011, 2015).

Expectations from an anthropomorphic robot

Another possible reason for the preference for the VR is that the expectations from the robot were higher due to its anthropomorphic properties. It has been suggested that the acceptance of a robot diminishes when it does not meet expectations (Lohse, 2011). For example, in the current study, about 15% of the participants complained that the robot was too slow and took too many breaks ("why is he getting tired so quickly?"), while no such comments were made regarding the VR system, even though the number and duration of breaks were identical across the two platforms. These comments are in line with previous studies showing that the timing of the robot's actions affects the users' perception of it (Eizicovits et al., 2018; Feingold Polak et al., 2018; Langer and Levy-Tzedek, 2020).

Projected preferences for long-term training

In contrast to the users' responses regarding the short-term training they underwent, when asked about their platform preference for long-term training, there was no clear preference for one system over the other. In other words, the preference for the exciting virtual experience offered by the VR in a single session diminished when asked to project their preferences to the long term, perhaps as considerations of physical comfort and difficulty in concentrating weigh in (see below). It may also be that, in the long term, people assign more weight to the social nature of the interaction with the robot. Even so, responses were split in half on the question of which platform they preferred to train with in the long term, and there was no preference for one platform over the other. It has previously been suggested that the perception of *social presence* increases in long-term interactions with social robots (Lee et al., 2005). It is possible that multiple sessions with the robot would lead to a different distribution of preferences. Another possible explanation for the equal preferences for the VR and the robot in the long term is the level of comfort of using the VR software. About a third of the participants found the diversity of the visual elements presented in the VR to be distracting and confusing. In other words, the variety of visual details, which contributed to the creation of an immersive VR environment and a sense of spatial presence, may also have led participants to not favor VR in the long term. Lastly, the physical comfort of the VR setup may also have played a role in the diminished preference for VR in the long term. Participants reported that they found the interaction with the robot to be more comfortable than with the VR. It is possible that the discomfort (e.g., cybersickness, heavy headset, etc.) that VR setups sometimes elicit (Moss and Muth, 2011) may discourage participants from choosing to use it over multiple sessions. Importantly, we did not directly test long-term training with the system, but rather asked participants to project their expected preferences, had training been over an extended period of days. Future studies should clearly identify the factors that contribute to engagement in a one-time session versus over multiple sessions.

Future directions

It may be that alternating between platforms – a suggestion that was repeatedly brought up by participants in the current study – would be successful in maintaining user engagement, interest, and involvement over the long term. An additional solution that may build on the benefits of both platforms is to use a hybrid technology, such as augmented reality (AR), a technology that merges virtual elements with the actual reality and is used in various fields including rehabilitation (Desai et al., 2016), education (Akçayır and Akçayır, 2017), and industry (Fraga-Lamas et al., 2018). By using AR technology, users may benefit from the physical presence of a social entity, while interacting with artificial stimuli that can enhance their arousal and feeling of immersion during the experience.

The participants expressed positive and enthusiastic responses regarding both platforms, when tested in the laboratory, using high-end professional equipment. This encourages future work to develop adjustments to enable home-based cognitive training, using these or similar technologies. With the advent of technology towards home-based training (Payne and Stine-Morrow, 2017; Stuifbergen et al., 2011; Wilms, 2020), it is important to address the challenges that this setting presents. A major challenge in home-based training is sustaining engagement over the long-term, especially when it involves independent use (Lampit et al., 2014). Here, we explored ways to increase engagement and motivation by diversifying approaches to the same task, incorporating gamification and creating presence-enhancing experiences. Future work can focus on other aspects that may motivate users to engage in home-based cognitive training delivered by these platforms, such as personalization (Shatil et al., 2010), performance monitoring (Sailer et al., 2014), and collaboration with other users (Arlati et al., 2019). The use of these technologies in home-based settings raises additional issues that will need to be addressed in future studies, such as accessibility to different populations, convenient operation, affordable equipment, and safety.

Study limitations

The results of the study are based on a particular robot embodiment, with the human-like "personality" we designed for it, and on the particular head-mounted display we used, with the set of virtual scenes we designed for it. We designed a virtual-reality training platform that creates a strong sense of spatial presence, and an interaction with a social robot that creates a strong sense of social presence; in an extension of this work, it would be possible to add, for example, a social component to the virtual setting and test how that might modify the users' preferences. Therefore, additional comparisons, using other robots and

virtual environments, with varying levels of spatial and social presence for each, are required to generalize the findings. It would be helpful, in future studies, to compare the users' engagement levels using these platforms with more traditional cognitive training settings, to assess the potential benefits of using these technological platforms. Furthermore, it would be instructive to test these different platforms in other frameworks besides cognitive training, such as in the context of physical therapy, rehabilitation, and education. The results reported here are based on a convenience sample of 64 individuals, tested in a lab setting. A larger group should be tested "in the wild" to verify the generalization of these results beyond the tested population and the settings of the current experiment. A long-term training program would be necessary in order to establish how users' preferences change over time.

5. Conclusions

We present here the design, implementation, and comparison of two technological platforms for use in cognitive training: a social robot and immersive virtual reality. Our design focused on "wrapping" traditional and common cognitive tasks in gamified settings in order to elicit engagement and motivation for training. This is the first study, to the best of our knowledge, to integrate classical cognitive-training tasks (the N-back and Task Switching) into these two technological platforms – immersive VR and a SAR. The design we present can be adapted per user to account for differences in personal preferences across users, and for changes in these preferences over time. We tested how each platform was perceived by users, whether there is a preference to use one of the platforms over the other for training in the short and the long term, and whether the preference correlated with interpersonal differences such as age, gender, personality traits, and empathic tendencies. We found that participants perceived the robot as a social empathic entity with human-like qualities and that they felt present in the VR experience, which elicited interest and excitement. We also found they preferred the VR platform across interpersonal differences in the short term, but the projected preference diminished in the long term, where the projected preference rates for both platforms were identical. In contrast to previous works that showed a tendency to prefer interaction with physically embodied social robots over virtual experiences, participants here preferred *spatial presence* in the VR over *social presence* with a physically present social robot for short-term cognitive training. The equal projected preference in the long term suggests that the effects of *social* and *spatial presence* are equally important for long-term use and that a combination of platforms may yield engagement over long-term training.

As older adults worldwide are not able to access community-based forms of cognitive training due to the global COVID-19 pandemic, it is of high importance to deploy home-based training systems that will keep them engaged and motivated in the long term.

Authors contribution

O.C. and S.L. designed the experiment. O.C. implemented the experimental design on the robotic and the VR platforms, carried out the experiment, and analyzed the data. O.C. and S.L. interpreted the results and wrote the manuscript. S.L. supervised the project and obtained the funding for it.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijhcs.2022.102775.

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